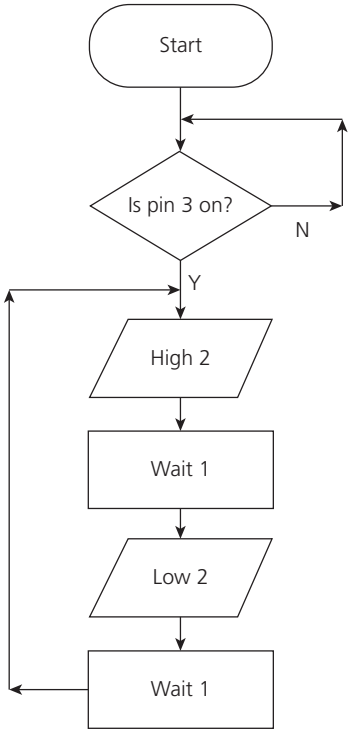


Figure 1.23 A program that uses time delays and loops to create a continuous pulsing sequence

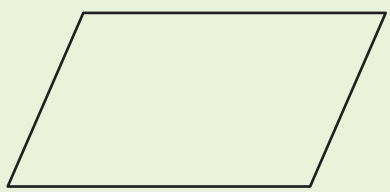
Exam tip

Make sure you can create flowchart programs that can read inputs, switch outputs on/off and produce more complex process functions, such as time delays and counting.

Now test yourself

TESTED

1 What does the following flowchart symbol represent? (1)



2 Draw the flowchart symbol for a process command. (1)

3 Explain how a microcontroller could be programmed to read an analogue value from a sensor. (3)

4 Write a microcontroller flowchart program that would turn an output connected to pin 4 'on' for two seconds when a switch connected to pin 3 is pressed, and then 'off' again. (6)

1.8 Ferrous and non-ferrous metals

1.8.1 Ferrous metals

REVISED

- **Ferrous metals** contain iron so they corrode quickly.
- They are magnetic and corrode quickly if not treated with a suitable surface finish.

Ferrous metals: metals that contain iron.

Mild steel

- Mild steel is also known as low-carbon steel because it has the lowest carbon content of the different steel types.
- Mild steel contains around 99 per cent iron and a maximum of 0.25 per cent carbon.

- Other elements such as manganese, silicon and sulphur make up the remaining fraction.
- Mild steel is relatively affordable compared to other types of steel.
- It is easy to machine and weld.
- Common uses are structural steel in buildings, signs, car and vehicle bodies and panels, and wire.

Exam tip

Remember that all steels are actually alloys as they are a mixture of iron, carbon and other elements.

Stainless steel

- Stainless steel contains iron and between 15 and 30 per cent of other metals (chromium and nickel).
- It has a high resistance to corrosion compared to other types of steel.
- It is tough and ductile with high strength and temperature resistance.
- Common uses are medical instruments, kitchen utensils and appliances, vehicle exhaust systems, hardware and structural/architectural items.

Cast iron

- Cast iron is a mixture of iron and two per cent or more of carbon.
- It is heavy and has good compressive strength but is quite **brittle** and can crack or shatter under impact.
- It has a low melting point and is more resistant to wear and corrosion than mild steel.
- Common uses are pipework, engines, cooking utensils, and garden ornaments and furniture.

Brittleness: hard but easily broken or shattered.

1.8.2 Non-ferrous metals

REVISED

- **Non-ferrous metals** do not contain iron.
- They are much more corrosion resistant than ferrous metals.
- They are generally more expensive than ferrous metals.

Non-ferrous metals: metals that do not contain iron.

Aluminium

- Aluminium has a high strength-to-weight ratio as it is lightweight.
- It has a shiny, attractive appearance that requires no additional finish.
- It is easy to cut, shape and weld.
- Common uses are drinks cans, foils, window frames and aeroplane parts.

Copper

- Copper is an excellent electrical and thermal conductor that does not tarnish or corrode easily.
- It is tough but ductile, so it can be easily shaped or drawn into a wire.
- Copper can be joined easily by soldering or brazing.
- It can be polished to an attractive appearance and golden colour.
- Common uses are water pipes, electrical cables, jewellery, statues and outdoor decorative features.

Brass

- Brass is an alloy of copper and zinc.
- Alloys are metals made by combining two or more metals and other elements together to improve their properties.
- Brass has good resistance to wear and corrosion, and has an attractive appearance.
- Common uses for brass are nuts, bolts and screws, door furniture and bathroom fittings.



Figure 1.24 Brass is an alloy made from copper and zinc

Typical mistake

Unlike gold and silver which are non-ferrous metals, brass is an alloy and not a 'pure' metal.

1.8.3 Properties

REVISED

The properties of different metals are the main factors that will influence their selection and use. The general properties of metals are:

- **high melting point**: resistant to very high temperatures
- **finish**: can be polished to an attractive shine
- **conductive**: allows electricity and heat to pass through easily
- **malleable**: can be hammered and bent into shape without breaking
- **ductile**: can be stretched into a thin wire without becoming weaker or brittle in the process
- **hardness**: resistant to impact and abrasion.

Conductive: allows electricity and heat to pass through easily.

Malleable: can be hammered and bent into shape without breaking.

Ductile: can be stretched into a thin wire without becoming weaker or brittle in the process.

Now test yourself

TESTED

1 Name the **two** differences between ferrous and non-ferrous metals. (2)
2 What is an alloy? (2)
3 All metals are good conductors. What does this mean? (2)

1.9 Papers and boards

1.9.1 Paper

REVISED

- Papers and boards come in a wide range of different thicknesses, sizes and types.
- The thickness of paper is known as its 'weight' and is measured in **grams per square metre (gsm)**.

Grams per square metre (gsm): the weight in grams of one square metre of paper.

Copier paper

- Copier paper is used extensively for printing, photocopying and general office purposes.
- It is approximately 80 gsm.
- It has a smooth surface making it ideal for most printers and photocopiers.

Exam tip

Papers heavier than 170 gsm are classified as boards.

Cartridge paper

- Cartridge paper is available in different weights between 80 and 140 gsm.
- It is thicker and more expensive than layout and copier paper.
- It has a slightly textured surface and is slightly creamier in colour.
- Cartridge paper is used by artists for sketching, drawing and painting as it has an ideal surface for pencil, crayons, pastels, watercolour paints, inks and gouache.

Tracing paper

- Tracing paper is used for making copies of drawings and fine details.
- It is thin and very transparent.
- It is also hardwearing and strong despite its lack of thickness.
- Mistakes made in pen can be scratched off using a sharp blade.

1.9.2 Board

REVISED

- Boards are usually classified by thickness as well as by weight.
- The thickness of board is measured in **microns**; a micron is one-thousandth of a millimetre.

Folding boxboard

- Folding boxboard is made up of three layers of card.
- The card layers have different qualities and properties: the two outer layers are high quality with a very smooth surface; the middle layer is lower quality but has high bulk and **stiffness**.
- A chemical coating on the top outer layer makes it ideal for high-quality printing and foiling.
- Folding boxboard is used for health and beauty products, frozen and chilled foods, pharmaceutical products and confectionaries.

Stiffness: resistance to bending and flexing.

Corrugated cardboard

- Corrugated cardboard is a strong but lightweight type of card.
- It is made from two layers of card with a fluted sheet in between.
- The fluted construction makes it stiff and difficult to bend or fold, especially across the flutes.
- Corrugated card can absorb knocks and bumps, and has good heat-insulating properties.
- Common uses are packaging fragile or delicate items, and takeaway foods, for example pizza boxes.

Solid white board

- Solid white board (also known as mounting board) is a rigid type of card with a thickness of around 1.4 mm (1400 microns) and a smooth surface.
- It is available in different colours but white and black are the most commonly available.
- Mounting board is often used for picture framing mounts and architectural modelling.

1.9.3 Properties

REVISED

- **Flexibility** is the paper's ability to withstand multiple folds before it breaks. This is important in printed products that may be folded and unfolded numerous times during use, such as books, maps and cartons.
- Paper made with virgin pulp, which has longer fibres, has a higher flexibility than recycled paper.
- **Printability** is how easy the paper can be printed on.
- The smoother and flatter the paper's finish, the easier it is to print on and the higher quality the print finish will be.
- **Biodegradability** means that the paper or board can break down and decompose without harming the environment.
- Normal paper and board will biodegrade quickly. Coatings on products such as coffee cups to make them waterproof significantly affect the length of time it takes the paper and board to decompose.

Flexibility: ability to bend and curve evenly and easily.

Printability: how easily a material can be printed on.

Typical mistake

Paper is not generally considered to be a strong material. However, for its weight it is a very strong material.

Now test yourself

TESTED

1 Name **one** use for tracing paper. (1)
2 Explain **two** reasons why corrugated cardboard is ideal for takeaway food packaging. (2)
3 Describe **one** use for mounting board. (2)

1.10 Thermoforming and thermosetting polymers

1.10.1 Thermoforming polymers

REVISED

- **Thermoforming polymers** soften when heated and can be moulded into shape.
- When the polymer cools it hardens and retains the shape.
- If the polymer is reheated it will try to return to its original shape.
- Thermoforming polymers can be reheated and reshaped numerous times.

Thermoforming polymer: a polymer that can be reheated and reshaped a number of times.

Acrylic

- Acrylic is a rigid, hard polymer usually in the form of a sheet.
- It is available in clear or a wide range of solid and translucent colours.
- Acrylic is easy to cut, bend, shape and polish.
- It has good **impact resistance** but is easily scratched.
- Common uses include as a substitute for glass, illuminated signs, aircraft canopies and car light clusters.

Typical mistake

Many students give acrylic as a polymer used for vacuum forming. Although it is easy to shape and bend using a strip heater, it is not suitable for vacuum forming.

Impact resistance: resistance to knocks and impacts.

High-impact polystyrene

- High-impact polystyrene (HIPS) is a thin, flexible and lightweight sheet polymer available in a wide range of colours.
- It is inexpensive and easy to cut, shape, bend and mould.
- HIPS is a popular choice for vacuum forming.
- Common uses for HIPS include disposable plates and cups, model kits, food containers and trays.

Biodegradable polymers

- Biodegradable polymers, or **biopolymers**, are produced from renewable raw materials such as corn starch, plant sugars and vegetable oil.
- They are environmentally friendly as they are compostable and biodegradable.
- Biopol® is a type of biopolymer. It is lightweight, hard and non-toxic.
- Common uses for biodegradable polymers include medical products such as dissolving thread for stitching wounds, bin bags, bottles and cups, and food packaging.

Biopolymer: a biodegradable polymer made from a plant source, such as corn starch or sugar beet.

1.10.2 Thermosetting polymers

REVISED

- **Thermosetting polymers** soften when heated and can be moulded into shape.
- During heating, a chemical change takes place.
- Once the thermosetting polymer has cooled, it hardens and sets in shape.
- Once set, the thermosetting polymer cannot be reheated and reshaped.
- If reheated, the thermosetting polymer will melt and burn.
- Thermosetting polymers cannot be recycled.

Thermosetting polymer: a polymer that can be heated and shaped only once.

Exam tip

Apart from biopolymers, polymers are not considered to be environmentally friendly materials and their use in modern society is causing considerable damage to the environment.

Polyester resin

- On its own, polyester resin is hard but very brittle and easy to snap.
- When combined and reinforced with fibreglass, it becomes GRP (glass reinforced plastic).
- GRP is very tough and has high impact resistance. It is also lightweight, waterproof and relatively easy to repair if damaged.
- Common uses for GRP include car body panels, sailing boats, canoes, corrugated roofing sheet, and electrical enclosures.

Urea formaldehyde

- Urea formaldehyde is a rigid, hard but brittle polymer.
- It is available in a wide range of colours.
- It has excellent heat resistance and is a good electrical **insulator**.
- Common uses for urea formaldehyde include electrical fittings (light switches, plug sockets), domestic appliance parts, laminate sheeting and wood glue.

Insulator: a material that does not conduct heat or electricity easily.

1.10.3 Properties

REVISED

The general properties of polymers are:

- **thermal insulator**: does not conduct heat easily
- **electrical insulator**: does not conduct electricity
- **toughness**: resistant to impact and wear.

Now test yourself

TESTED

1 Explain why thermoforming polymers are considered to be more environmentally friendly than thermosetting polymers. (2)
2 Give **two** reasons why polymers are often used for housing electrical equipment. (2)
3 Name a common use for HIPS. (1)

1.11 Natural, synthetic, blended and mixed fibres, and woven, non-woven and knitted textiles

- Fibres are very fine, hair-like structures that are spun (or twisted) together to make **yarn**.
- Yarns are woven or knitted together to create textile fabrics.

Yarn: spun threads used for knitting, sewing and weaving.

1.11.1 Natural fibres

REVISED

- Natural fibres come from natural sources.
- They are sustainable and biodegradable.

Table 1.10 Sources, properties and uses of natural fibres

Fibre	Source	Properties	Uses
Wool	Animal: sheep fleece	Warm, **absorbent**, low flammability, good **elasticity**, crease resistant	Warm outerwear (coats and jackets); knitwear; soft furnishings, carpets and blankets
Cotton	Vegetable: cotton plant/ seed pod	Absorbent, strong, cool to wear, hardwearing, creases easily, smooth, easy to care for, flammable, can shrink	Clothing, bedlinen, towels, sewing and knitting threads, soft furnishings

Absorbency: ability to soak up moisture.

Elasticity: ability to stretch and return to shape.

1.11.2 Synthetic fibres

REVISED

- Synthetic, or manufactured, fibres are artificial.
- Advantage: they can be manufactured to specific requirements.
- Disadvantage: they are not biodegradable and come from unsustainable sources.

Table 1.11 Sources, properties and uses of synthetic fibres

Fibre	Source	Properties	Uses
Polyester	Fossil fuels: coal/ petroleum	Strong when wet and dry, flame resistant, thermoplastic, hardwearing, poor absorbency	A versatile fibre used throughout textiles
Acrylic	Fossil fuels: petroleum	Strong except when wet, thermoplastic, burns slowly then melts, poor absorbency	Knitwear and some knitted fabrics, fake fur products including toys, upholstery and carpets

Exam tip

Look at the labels of different products. Try to identify the fibre used and work out what properties the fibre has that will allow the product to function as intended. The more you look at, the better your understanding of material selection will be.

1.11.3 Woven textiles

REVISED

- Woven fabrics consist of **warp** and **weft** yarns in an under/over configuration.
- Different **weave** creates fabrics with different **textures**, patterns and strength.
- **Plain weave** is the most basic weave. It is stable, strong and gives an even surface on both sides of the fabric. An example is calico.
- **Twill weave** gives a diagonal pattern and adds strength to the fabric. An example is denim.

Warp: yarn following the length of a fabric.

Weft: yarn running horizontally across a fabric.

Weave: the way yarns are laid in a pattern/formation in the construction of fabric.

Texture: the feel and finish of a material.

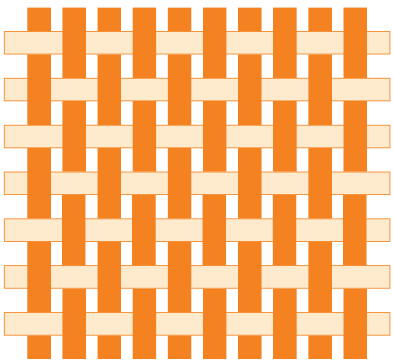


Figure 1.25 A plain weave structure

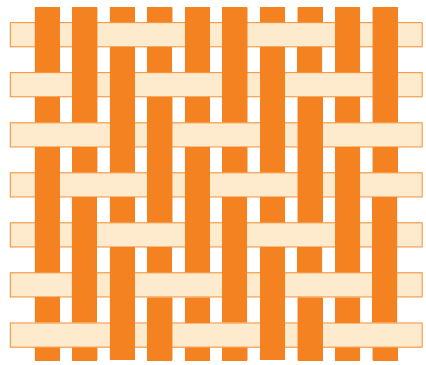


Figure 1.26 A twill weave structure

Typical mistake

Don't confuse warp and weft yarns used in fabric construction. Warp run the length of the fabric and weft run across the width of fabric from left to right.

1.11.4 Non-woven textiles

REVISED

Felted

- Felted fabrics are produced by applying moisture, heat, pressure and friction to a web of fibres, which causes the fibres to bind together. Wool and acrylic fibres are commonly used in this process.
- Felt is used in craft projects, accessories such as hats, and on the surface of snooker tables.

Bonded

- Non-woven or bonded textiles are **constructed** from a web of fibres, held together by **adhesive** or stitching.
- Most non-woven or bonded fabrics are not strong enough to be made into garments but are often used to reinforce woven and knitted fabrics.
- They are cheap to produce and are often used to make disposable products such as surgical gowns and masks, cleaning wipes and disposable protective suits.

Constructed: the way something has been made.

Adhesives: glues.

1.11.5 Knitted textiles

REVISED

- Knitted fabrics are made by creating a series of loops in the yarns that interlock together.
- There are two types of knitted fabrics: **warp knitting** (interlinking vertically) and **weft knitting** (interlinking horizontally).
- Knitted fabric is easy to stretch and warm to wear as the loops trap body heat.

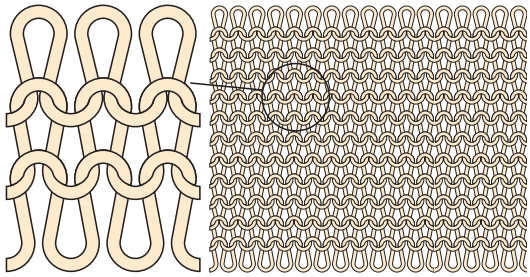


Figure 1.27 Warp knitting consists of multiple yarns that loop together and interlock vertically

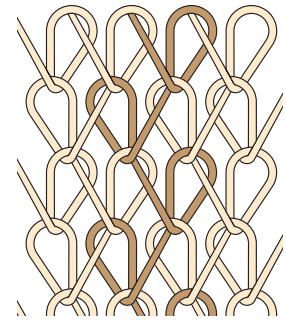


Figure 1.28 Weft knitting consists of one yarn with a series of loops that interlock horizontally

1.11.6 Properties

REVISED

Fibres have different properties and characteristics depending on the source of fibre and how the fabric has been constructed. This affects what it can be used for. Table 1.12 shows the performance properties of different fibres. The higher the star rating, the greater the property in the stated fibre.

Table 1.12 Performance properties of fibres

	Wool	Cotton	Polyester	Acrylic
Elasticity	****	*	****	***
Resilience	****	*	****	***
Durability	**	**	****	***

Now test yourself

TESTED

1 Name **one** natural fibre and **one** synthetic fibre. (2)
2 Explain how the type of weave in fabric construction affects the fabric. (2)
3 State **one** characteristic of knitted fabric. (1)
4 Explain why bonded fabrics are often used in disposable products. (2)
5 Describe **two** reasons why natural fibres are considered more environmentally friendly. (2)

1.12 Natural and manufactured timbers

- **Natural timber** comes from trees that are cut down into logs and then sawn into planks.
- **Manufactured timber** is made from recycled wood chips and fibres.

Natural timber: timber that comes from trees.

Manufactured timber: a human-made board made from recycled timber.

1.12.1 Natural timbers – hardwoods

REVISED

- **Hardwoods** come from deciduous trees that shed their leaves every year.
- Hardwood trees grow slowly and can take hundreds of years to grow fully.
- They have thick trunks with branches at the top.
- Hardwoods have a close **grain** and tend to be denser and heavier than softwoods.

Hardwood: timber that comes from deciduous trees.

Grain: the longitudinal arrangement of wood fibres.

Oak

- Oak is a heavy, hard and tough timber.
- It has an open grain and an attractive appearance.
- Oak is good for outdoor use as it is durable and can withstand the elements well compared to other woods.
- Common uses for oak include garden furniture, doors, floors and high-end furniture.

Mahogany

- Mahogany is a dark coloured close-grained timber.
- It is easy to cut and shape, is available in wide planks and polishes well.
- Common uses for mahogany include furniture, shop fittings, boat building and doors.

Beech

- Beech is a very tough, hard timber with a close and straight grain.
- It is hardwearing, easy to work and polishes well.
- It has a light brown colour and can be prone to warping.
- Common uses for beech include furniture, toys and wooden tools.

Balsa

- Balsa is a soft and lightweight timber with a coarse and open grain.
- It is very easy to shape, sand, glue and paint.
- Common uses for balsa include model making, packing cases, surfboards and fishing floats.

Typical mistake

Balsa is a hardwood, despite it being a very soft, lightweight and easy to shape.

1.12.2 Natural timbers – softwoods

REVISED

- **Softwoods** come from coniferous (evergreen) trees that have needles instead of leaves.
- Softwood trees keep their needles year-round and grow much faster than hardwood trees.
- They grow tall and straight, and have lots of branches all the way up the trunk.
- Softwood has more knots than hardwood.

Softwood: timber that comes from coniferous trees.

Pine

- Pine is the most widely available natural timber and is relatively cheap compared to other timbers.
- It has a straight grain and is easy to cut and shape.
- Pine has knots along the grain of the wood. They can weaken its strength but give an attractive appearance when finished.
- Common uses for pine include interior construction work, boxes and crates, flooring and paper (pulpwood).

Cedar

- Cedar is a lightweight, soft timber with a straight grain.
- It is knot free and very durable.
- Cedar is resistant to rot, weather and insect attack.
- Common uses for cedar include outside joinery, building cladding, bathroom and kitchen furniture, and wall panels.

Exam tip

Hardwoods are generally heavier, harder, tougher and more durable than softwoods.

1.12.3 Manufactured timbers

REVISED

- Manufactured boards can be made from hardwood and softwood.
- They are made by gluing and compressing wood fibres or layers together.
- They are made in large sheets that are easy to work with.
- Manufactured boards are generally cheaper than 'real' or natural wood.



Figure 1.29 Plywood is made from thin layers of timber placed with alternating grain direction

Plywood

- Plywood is made by gluing together thin layers of timber with alternating direction of grain.
- It is strong and tough due to its layered construction, with a high strength-to-weight ratio.
- Plywood is easy to cut and shape but can splinter easily.
- It is relatively stable under changes in temperature and moisture.
- Common uses for plywood include structural panelling in building construction and furniture making.

Medium-density fibreboard

- Medium-density fibreboard (MDF) is made by gluing and compressing wood fibres together into sheets.
- It has a smooth, even surface but chips and damages easily.
- It is easy to cut, shape, paint, glue and stain.
- MDF has poor moisture resistance and degrades quickly when wet.
- Common uses include cheap flat-pack furniture, interior panelling and interior doors.

1.12.4 Properties

REVISED

Hardness

- **Hardness** is the resistance of the wood to denting and wear.
- It varies depending on the direction of the wood grain.
- Hardness of wood is tested by dropping a ball bearing onto it from a set height. This is called the Janka hardness test.

Hardness: resistance to impact and abrasion.

Toughness

- Toughness is the amount that timber can be bent without breaking or splitting.
- Toughness is affected by the timber's internal cell structure.
- Most hardwoods have interlocking cell structures and are generally tougher than softwoods.

Durability

- **Durability** is the wood's ability to resist decay from exposure to the elements and attack from insects.
- Hardwoods such as teak are the most durable woods and can last for many years when exposed to the elements.
- Some types of bamboo have a durability of only a few months.

Durability: resistance to wear, long lasting.

Now test yourself

TESTED

1 Why is plywood a much stronger manufactured board than MDF? (2)
2 Name **one** advantage and **one** disadvantage of knots in pine. (4)
3 Explain **one** major difference between hardwood and softwood trees. (2)

1.13 Contexts which inform design outcomes

1.13.1 Discriminating between materials, components and manufacturing processes

REVISED

Properties may change depending on things such as temperature or the shape and direction of the material. Other properties will remain the same regardless of the environment or conditions.

Mechanical properties include:

- compressive strength – resistance to a pushing force
- tensile strength – resistance to a pulling force
- durability – resistance to wear, long lasting
- hardness – resistance to abrasion
- stiffness – resistance to bending and flexing
- elasticity – ability to stretch and return to shape
- impact resistance – resistance to knocks and impacts
- brittleness – hard but easily broken or shattered
- water resistance – resistance to water or moisture
- absorbency – ability to soak up moisture.

Manufacturing properties include:

- machinability – ability to be easily milled and drilled
- workability – ability to be placed and compacted
- ductility – ability to stretch, not break under stress or impact
- malleability – ability to be hammered, pressed or rolled without breaking.

Other properties include:

- acoustic – acoustic absorption
- chemical – corrosion resistance
- electrical – conductivity, resistance and so on
- thermal – conductivity, flash point and so on
- magnetic – attractive to magnets.

Typical mistake

Malleability and ductility are similar properties and often confused. Ductility relates to the ability of a material to stretch without damage. Malleability relates to the ability of a material to deform under compression.

Exam tip

By learning the properties of materials, you can understand why they have been used to make certain products. This will help you to answer questions in the exam about individual products and their fitness for purpose.

Advantages and disadvantages

- The choice of materials impacts the success of the final product.
- The advantages of a particular material may make it an obvious choice for a product, but the designer must also consider disadvantages, for example a thermosetting polymer may have the best properties for a product but it is not recyclable. Therefore, the designer may choose a thermoforming polymer with properties as close to the thermosetting one as possible.

Justification of the choice

- The designer must make informed choices based on these characteristics and other factors, such as:
  - cost – is there another material available that may cost less?
  - availability – is the material in good supply?
  - sustainability – is the material sustainable?
  - environmental impact – can the material be recycled?
  - social, cultural and moral considerations – will the use of this material have a negative impact?
- When designing an electronic system, the designer must consider the range of components available and decide on the most suitable type for the application.

Now test yourself

TESTED

1 State **four** important mechanical properties of materials. (4)
2 State **two** factors that can influence and affect the properties of materials. (2)

1.14 Environmental, social and economic challenges

1.14.1 Respect for different social, ethnic and economic groups

REVISED

Positive and negative impacts of products

- Products can have a negative impact on society that the designer may not have foreseen.
- For example, leaflets and flyers can be used to communicate useful information to people about services in the area, however this can lead to unwanted junk mail being delivered, urging people to buy things they don't want.
- While it is not always easy to predict the effects of a product, designers must try to consider the possible negative implications and decide whether the product is worthwhile.